

Resume — Tims Pečerskis, MSc

Latest version: peersky.xyz/resume · PDF: peersky.xyz/resume.pdf

Systems engineer, algorithm designer, and product builder. 15+ years designing novel algorithms under real-time, resource-constrained conditions and shipping them as production systems — **from bare metal signal processing through distributed GPU orchestration**. Proven ability to lead cross-discipline teams, manage multi-client product portfolios, own end-to-end product lifecycles from user research through delivery, and drive technical sales from presales through production support. Published researcher, patent author, and conference speaker with a track record of consistently outperforming project timelines.

peersky.xyz · github.com/peersky · linkedin.com/in/peersky · t@peersky.xyz

Domain Expertise

Product & Project Leadership — Product design · product strategy · user research · roadmap ownership · reliability & scalability planning · SDLC and hardware lifecycle management · agile · compliance & certification · cross-functional team leadership · hiring & org scaling · performance engineering · reliability engineering · portfolio management · multi-client delivery.

Algorithm Design & Systems Engineering — Real-time signal processing · low-latency systems · deterministic processing · hardware-software co-design · latency profiling · machine learning · protocol security · software engineering · electrodynamics · data analysis · R&D · patent development and IP strategy.

Business Operations — Budget ownership · process optimisation · vendor management · stakeholder communication · compliance framework navigation.

Growth — Technical sales · presales · account management · solutions architecture · developer advocacy · developer experience design · product-led growth · public speaking · technical writing.

Professional Experience

Founder — [Peeramid Labs](#)

November 2024 — Present

Lean team of three, experimenting across AI infrastructure, blockchain security, and privacy.

Skills: Rust · C · CUDA · Python (pandas, scipy, sklearn) · Huggingface · vLLM · SGLang · NATS.io · Docker · Redis · Postgres · GCP · TS/JS (React, Node) · LaTeX · Solidity · EVM (Arbitrum, Ethereum, Reth, Celo, Optimism, zkSync, Base, smart accounts, MEV) · Circom · Foundry · wagmi · alloy-rs · Zama FHE · Web3 security · protocol engineering · game-theoretic and financial modelling · P2P networking · business development · investor relations · product strategy for compliance and regulation.

Achievements:

- Shipped three ML products end-to-end — defining product scope, designing algorithms, and leading implementation: [AI code-slop firewall](#), smart-contract audit tool (**83% precision**), and a health-prediction engine ($R^2 = 0.52$ on held-out subjects).
- Designed **Noolog** end-to-end as a product — from research paper to shipped platform. Sovereign agent-to-agent reasoning network implementing NSED: agents propose, cross-evaluate, and converge with full audit trace. Event-driven low-latency Rust system on NATS JetStream — immutable per-round traces, live SSE streaming, runtime human oversight and kill switch. Led a team of three across architecture, compliance, and product design. **84% on AIME 2025** with three 8–20B local models, matching DeepSeek-R1 on **4x less VRAM**.
- Authored [Mixture-of-Models: Unifying Heterogeneous Agents via N-Way Self-Evaluating Deliberation](#) (NSED, arXiv:2601.16863) — a distributed GPU neural architecture for heterogeneous model orchestration delivering **4x CapEx reduction** or **30% accuracy boost** on any base model benchmarked.
- Authored [BRO-AMM: A Bonded Reconciliation Oracle AMM as a Low-Risk DeFi Primitive](#) (preprint) — a Uniswap v4 hook that settles each swap's effect on liquidity providers after the block, using chain history and signed market prices. In simulation on six months of Ethereum and Arbitrum ETH/USDC data: LP losses to arbitrage down **68–79%** (Ethereum) and **49–55%** (Arbitrum),

mainnet LP net turning from loss to gain, settlement at **0.18x** the gas of the swaps it settles.

- Built the production Rust implementation of NSED on NATS JetStream. Open-sourced the A2A agent SDK [quorum-rs](#) on crates.io with docs served to humans and agents over MCP at [noolog.io](#).
- Designed the Noolog product experience: Rust-native desktop client, compliance-native architecture generating **EU AI Act / GDPR / SOC2** audit artifacts from deliberation traces, developer SDK, and MCP integration for agent-facing documentation.
- Filed **6 provisional patent applications across 5 families** — multi-agent AI coordination, novel AI architecture (zero prior art), entropy-based model reputation systems, cryptographic compliance infrastructure, and hardware-bound licensing.
- Identified and responsibly disclosed a critical vulnerability in **Ethereum EIP-7702** — [full write-up](#).

Lecturer, Blockchain Architecture & Programming — [Invisible Garden Foundation](#), US

October 2024 — Present

Skills: Teaching · blockchain architecture · EVM · Solidity · resource optimisation · BFT consensus · protocol design fundamentals.

Achievements:

- Tutored **100+ students** over two years in blockchain architecture and programming — EVM, Solidity, resource optimisation, BFT fault tolerance, and the basics of protocol design.

Solution Architect / Senior Solution Developer — [OpenZeppelin](#), UK

November 2022 — August 2024

Skills: AWS · KMS · Terraform · DataDog · Grafana · IaC · technical account management · pre-sales · solution engineering · solutions architecture · public speaking · incident response planning · audits · DevSecOps · security monitoring · governance monitoring · DeFi · Cosmos SDK · Solidity · Rust · wargames.

Achievements:

- Led **technical account management and pre-sales** for the customer portfolio — owning discovery, scoping, product-fit qualification, and post-sale delivery handoff. Translated customer security requirements into deployable product configurations on the Defender platform.
- Served as technical advisor on the BIS [Project Mariana](#), coordinating across institutional stakeholders to contribute to the successful launch of the world's largest cross-institutional banking CBDC prototype.
- Co-authored the [EEA DeFi Risk Assessment Guidelines](#) — an industry-standard technical specification shaping how protocols assess and communicate risk.
- Led deployment of real-time monitoring and automated incident-response infrastructure on OpenZeppelin's IaC [Defender](#) platform, delivering near-100% continuous risk-factor monitoring coverage for customers.
- **Zero hacks on my customer portfolio during my shift.**

Product Engineer — [Moonstream](#) (formerly *Simiotics / Bugout*), California, US

June 2021 — August 2022

Skills: JS · TS · React · web3.js · ethers.js · Brownie · Figma · product design · UX research · big-data analysis.

Achievements:

- Shipped two products from zero in a fast-paced startup environment:
 - **Bugout** — developer-tooling analytics platform.
 - [Moonstream](#) — product design and UX research for drop-in game economy mechanics (lootboxes, crafting, achievements, leaderboards) aimed at quick developer integration. Platform processed **\$4B+ in transaction volume** in its first months post-launch.

Head of Engineering — SIA TeknoLogic, Latvia

February 2016 — November 2020

Owned the OEM/ODM product portfolio for all key accounts. Led cross-functional teams spanning hardware, firmware, certification, supply chain, and business development. Delivered **30+ distinct hardware/software products** end-to-end (concept → production) with near-100% customer retention.

Skills: C · C++ · Assembly · embedded Linux · Yocto · RTOS · Qt · Altium Designer · KiCad · high-speed digital / power / mixed / analog / PCI-E system design · schematic and PCB · BoM management · ARM-M and ARM-A · Nordic nRF · STM32 · IAR · GCC · CMake · EMI/EMC certification · DFM · supply-chain development · sales · business development · resource planning.

Achievements:

- **Erica Synths** — co-defined the product specification for the **Zen Delay** with Ninja Tune, translating artist requirements into technical constraints. Led cross-functional delivery across hardware, firmware, and manufacturing. Designed the DSP module — hardware and algorithm — with clock-level optimisations achieving a **30% BoM cost reduction** against competitors. Top reviews from [Coldcut](#), The Bug, Toro y Moi and others.
- **Erica Synths** — led mixed-signal system design for the **Black Dual VCF** around the Curtis CEM3320 filter chip re-issued as the **AS3320** — bridging digital control algorithms and analog signal path.
- **Erica Synths** — designed the **DIY Delay full kit** as a platform product anchoring Erica Synths' developer and hacker community — product strategy driving community growth.
- **Aretai Speakers** — owned the hardware product portfolio; **Aretai Contra received the Red Dot Award in 2022**. Hardware included precision electromechanical systems requiring tight tolerancing.
- **Coastal Amusements** — led the full product redesign of the amusement-machine motherboard, managing cross-functional delivery across schematic, PCB, firmware, and certification. Designed real-time multi-bus control system — **>200A load**, automotive-grade security over I²C, SPI and USB. Engineered as a modular product platform supporting arbitrary customer business objectives.
- **LeClanché** — delivered the **Battery Management System** power circuitry for high-reliability lithium battery packs.
- **Barix** — led development and full **EMI/EMC compliance certification** for the [Barix Exstreamer M400](#) audio-over-IP appliance.

Product Owner — [Edge Autonomy](#) (acquired by [Redwire](#), NYSE: RDW)

March 2015 — February 2016

Product Owner managing a cross-discipline team of **15 engineers** spanning electronics and mechanical. Owned the product roadmap, defined priorities, and set up the company's documentation and delivery infrastructure.

Skills: Radio engineering · product ownership · documentation · training · leadership · project management · resource planning.

Achievements:

- Increased operational range of video-streaming radio links from **5 km to 140 km+** by fixing design flaws in the existing platform. System operated under strict latency and throughput constraints for live video relay.
- Cleared a **2-year product backlog in a single quarter** by ruthlessly triaging scope and aligning engineering priorities with product roadmap.
- Outperformed the project roadmap by **3x**.

Microwave / R&D Engineer — [SAF Tehnika](#)

June 2011 — February 2015

Core contributor to the company's flagship [Integra](#) microwave radio. Combined algorithm design with RF systems engineering to create new product categories and reduce manufacturing costs.

Skills: Microwave engineering (5.6–60 GHz) · low-noise amplifiers · passive circuits · microstrip design · CST Studio · Matlab · RF synthesis (Genesys) · Altium Designer · documentation · LoRa · DFM.

Achievements:

- Designed and patented the core signal-processing algorithm for the **Spectrum Compact** — a world-first hand-held battery-powered spectrum analyzer that became SAF Tehnika's best-earning product. Real-time processing under strict power and size constraints.
- Core contributor to the **Integra** microwave radio backbone: designed microwave filters, orthomode transducer (OMT), and led R&D validation for orbital angular momentum antenna concepts.
- Designed microwave circuits for flagship products — ultra-wide-bandwidth **3-6 GHz amplifiers and LNAs**, mixers, polarisation splitters, microwave filters, waveguides, antennas. Typically outperformed timelines by **~2x**.
- Applied Monte-Carlo statistical modeling to manufacturing tolerance analysis, optimizing yield under tight RF performance constraints. Achieved **30%+ cost reduction**.

Patents & Publications

- **BRO-AMM: A Bonded Reconciliation Oracle AMM as a Low-Risk DeFi Primitive.** Preprint, 2026. doi.org/10.5281/zenodo.23241545
- **Mixture-of-Models: Unifying Heterogeneous Agents via N-Way Self-Evaluating Deliberation.** 2026. doi.org/10.5281/zenodo.18338777
- **EEA DeFi Risk Assessment Guidelines — Version 1.** EEA Technical Specification, 17 July 2024.
- **Spectrum Analyzer of Ultra-Low Intermediate Frequency.** LV 14855A, 2014, SAF Tehnika.
- **6 provisional patent applications filed across 5 families** — multi-agent AI coordination · novel AI architecture (zero prior art) · entropy-based model reputation · cryptographic compliance infrastructure · hardware-bound licensing.

Education

- **MSc, Electrical and Electronics Engineering** — [Riga Technical University](#).
- **BSc, Electrical and Electronics Engineering** — [Riga Technical University](#).

Languages

- English — Full Professional
- Latvian — Bilingual
- Russian — Bilingual
- Spanish — Limited
- Chinese — Elementary (studying)

After Work

Watersports (surfing, diving, freediving) · cycling · yoga · hiking · climbing · music (production, [SoundCloud](#)) · woodworking · retro / MMORPG gaming (retired WoW / L2 raid master).